DO NOW — RECALL

Trees and binary search trees

Name: _____ Class: _____ Date: _____

1. In chain A→B→C, which node follows B?

2. What does LIFO mean?

3. What must be true before binary search on an array?

4. Insert X between A and B: what should new X.next point to?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

C.

2. Model answer

Last In, First Out.

3. Model answer

The array must be sorted.

4. Model answer

B, before changing A.next to X.

Vocabulary — know these terms

root — Topmost tree node.

child — Node directly below a parent.

subtree — A node and all its descendants.

binary search tree — A binary tree ordered by values, with smaller keys to the left and greater to the right.

balanced — Height kept relatively small compared with node count.

NEW LEARNING

Read, annotate and explain

Start at root

Compare the new value 25 with root 40; it belongs in the left subtree.

Compare again

25 is greater than 20, so move right. The right child currently stores 30.

Insert without breaking the property

25 is smaller than 30; insert it as the left child of 30.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Insert 12 into BST with root 20, left 10 and right 30.

Predict

Compare 12 with 20 → go left to 10.

Confirm and explain

12 > 10, so 12 becomes the right child of 10.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

BST root 50, left child 25, right child 75: where does 60 go?

Question 2

Insert 15, 10, 20, 18 into an empty BST. Identify the parent of 18.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why a BST can have O(n) worst-case search time despite O(log n) average search on a reasonably balanced tree.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every binary search tree has logarithmic worst-case lookup time.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.